ECC-202

INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DIGITAL COMMUNICATION

Quiz, Spring Semester, 2025

Time Allowed: ONE hour **Total Marks: 15**

---

1. A signal $x(t) = N\cos((200+N)\pi t) + (N+5)\cos((200-N)\pi t)$ undergoes ideal sampling at a frequency of 150 samples per sec where N represents the last three digits of your enrollment number. Further, the sampled signal is filtered using a unit gain low pass filter (LPF) with Nyquist cut-off frequency.

   (a) Draw the amplitude spectra of the sampled signal. (1 mark)

   (b) Determine the frequency components present in the output of the LPF. (1 mark)

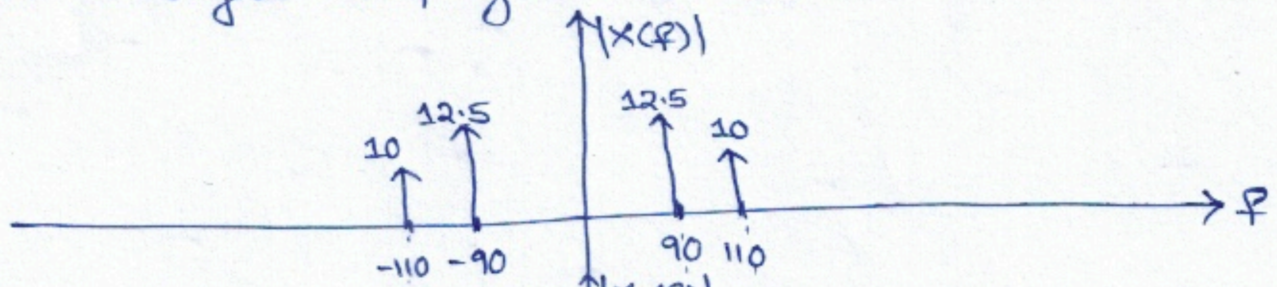
   (c) Write the time-domain expression for the output of the LPF. (2 marks)

Considering $N = 20$, $x(t) = 20\cos(220\pi t) + 25\cos(180\pi t)$

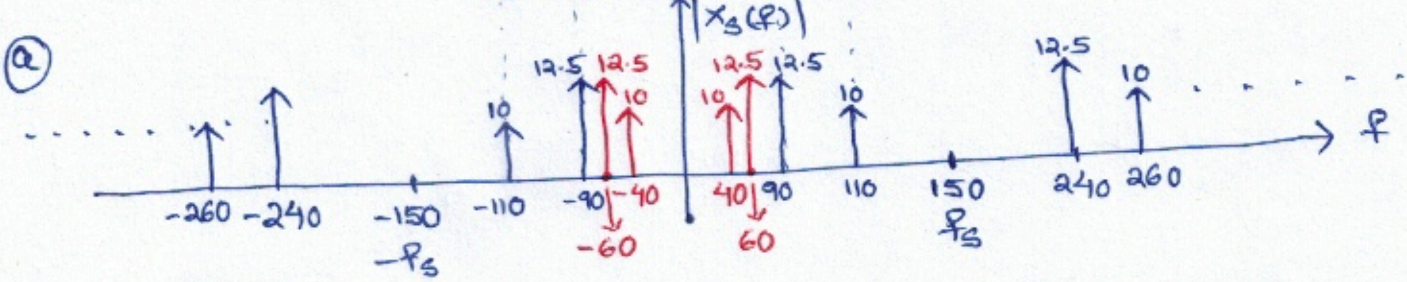
$$= 20\cos(2\pi\cdot 110\cdot t) + 25\cos(2\pi\cdot 90\cdot t)$$

$$\Rightarrow X(f) = 10\left[\delta(f+110) + \delta(f-110)\right] + 12.5\left[\delta(f+90) + \delta(f-90)\right]$$

$x(t)$ undergoes sampling at $f_s = 150$ Hz.



(a)



(b) Frequency components present in the output of LPF are

$$\underbrace{40\text{ Hz},\ 60\text{ Hz}}_{\text{Aliased frequencies}},\ \underbrace{90\text{ Hz},\ 110\text{ Hz}}_{\text{Original frequencies}}$$

(c) Output of LPF is $x_r(t) = 20\cos(2\pi\cdot 110t) + 25\cos(2\pi\cdot 90t) + 20\cos(2\pi\cdot 40t) + 25\cos(2\pi\cdot 60t)$

2. Consider a signal whose probability density function is shown in Figure 1. Consider uniform

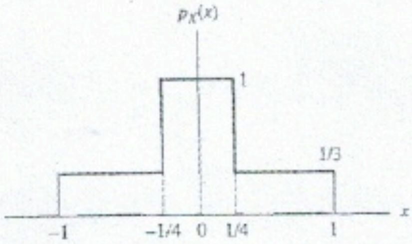


Figure 1

quantization with $L = 2^N$ where N represents the last three digits of your enrollment number. Determine the following:

(a) Quantization error variance (2 marks)

(b) Average signal to quantization noise ratio (1 mark)

(a) Quantization error variance:

$$\sigma_Q^2 = \frac{1}{3L^2} = \frac{1}{3(2^N)^2} = \frac{1}{3(2^{2N})}$$

(b) Average Signal Power:

$$\sigma_x^2 = 2\left[1 \times \int_0^{1/4} x^2\,dx + \frac{1}{3} \times \int_{1/4}^{1} x^2\,dx\right]$$

$$= 2\left[\frac{1}{192} + \frac{21}{192}\right] = 2 \times \frac{22}{192} = 0.2291$$

$$(SNR)_Q = \frac{0.2291}{1/(3 \times 2^{2N})} = 0.6875 \times 2^{2N}$$

3. A μ-law compander is defined by

$$y = \pm \frac{ln(1+\mu|x|)}{ln(1+\mu)}, \quad \text{for } |x| < 1$$

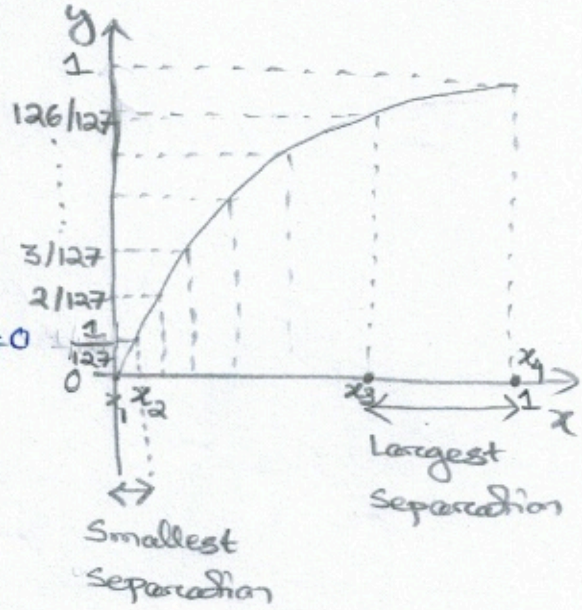
where x is the input and y is the output. The + sign is used when x is positive, and the - sign is used when x is negative. Consider the peak value of input is 10 volts, and the number of bits available for quantization are 8. If $\mu = N$ where N represents the last three digits of your enrollment number, determine the following:

(a) Smallest separation between levels (2 marks)

(b) Largest separation between levels (2 marks)

Considering the normalized input $|x| < 1$ i.e. $-1 < x < 1$, the Y-axis is uniformly quantized with step size

$$\frac{1}{\left(\frac{2^8}{2}\right) - 1} = \frac{1}{127}$$



Let $N = 255 \Rightarrow \mu = N = 255$

(a) Smallest separation occurs near to $x=0$ i.e. between $y_1 = 0$ and $y_2 = \frac{1}{127}$

$$\therefore \frac{\ln(1+255x_1)}{\ln(1+255)} = 0 \Rightarrow x_1 = 0$$

$$\frac{\ln(1+255x_2)}{\ln(1+255)} = \frac{1}{127} \Rightarrow x_2 = 1.75 \times 10^{-4}$$

$\therefore$ Smallest separation $= 10(x_2 - x_1) = 1.75 \times 10^{-3}$

(b) Largest separation occurs on x-axis between the levels $y_3 = \frac{126}{127}$ and $y_4 = 1$. Then,

$$\frac{\ln(1+255x_3)}{\ln(1+255)} = \frac{126}{127} \Rightarrow x_3 = 0.957$$

Similarly,

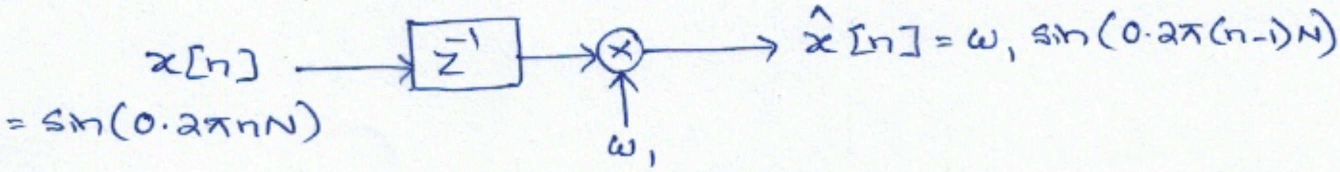
$$\frac{\ln(1+255x_4)}{\ln(1+255)} = 1 \Rightarrow x_4 = 1$$

$\therefore$ Largest separation $= 10(x_4 - x_3) = 10(1 - 0.957) = 0.43$

4. The input signal to a one-tap linear predictor is given by $x[n] = \sin(0.2\pi n N)$ where N represents the last three digits of your enrollment number. Determine the following:

(a) Tap weight for which the prediction error variance is minimized (1 mark)

(b) Minimum prediction error variance (1 mark)



Prediction error $e[n] = x[n] - \hat{x}[n]$

(a) Prediction error variance $J = E[e^2[n]]$

$$\Rightarrow J = E[x^2[n]] + E[\hat{x}^2[n]] - 2E[x[n]\hat{x}[n]]$$

$$\Rightarrow J = E[\sin^2(0.2\pi nN)] + \omega_1^2 E[\sin^2(0.2\pi(n-1)N] - 2\omega_1 E[\sin(0.2\pi nN)\sin(0.2\pi(n-1)N)]$$

$$= \frac{1}{2} + \omega_1^2 \times \frac{1}{2} + \omega_1 E[-2\sin(0.2\pi nN)\sin(0.2\pi(n-1)N)]$$

$$= \frac{1}{2} + \frac{1}{2}\omega_1^2 + \omega_1 E[\cos(0.4\pi nN - 0.2\pi N) - \cos(0.2\pi N)]$$

$$= \frac{1}{2} + \frac{1}{2}\omega_1^2 + \omega_1 \times 0 - \omega_1 \cos(0.2\pi N)$$

$$\frac{\partial J}{\partial \omega_1} = \omega_1 - \cos(0.2\pi N) = 0 \Rightarrow \omega_1 = \cos(0.2\pi N)$$

(b) $$J_{min} = \frac{1}{2} + \frac{1}{2}\cos^2(0.2\pi N) - \cos^2(0.2\pi N)$$

$$= \frac{1}{2} - \frac{1}{2}\cos^2(0.2\pi N) = \frac{1}{2}\sin^2(0.2\pi N)$$

5. A message signal $x(t) = 3\ \sin(Nt) + 4\ \sin(2Nt) + 4\ \sin(1.5Nt)$ is encoded using a delta modulator (DM) where N represents the last three digits of your enrollment number. Assuming no slope-overload distortion, determine the following:

(a) Quantization step-size (1 mark)

(b) Maximum output signal-to-quantization noise ratio (1 mark)

$$x(t) = 3 \sin(Nt) + 4 \sin(2Nt) + 4 \sin(1.5Nt)$$

(a) $$\frac{dx(t)}{dt} = 3N \cos(Nt) + 8N \cos(2Nt) + 6N \cos(1.5Nt)$$

$$\left|\frac{dx(t)}{dt}\right|_{max} = 3N + 8N + 6N = 17N$$

To avoid slope-overload distortion, $\frac{\Delta}{T_s} \geq \left|\frac{dx(t)}{dt}\right|_{max}$

$$\therefore \frac{\Delta}{T_s} \geq 17N \Rightarrow \Delta \geq 17NT_s \Rightarrow \Delta \geq \frac{17N}{f_s}$$

Maximum frequency in the signal $\omega_m = 2N \Rightarrow f_m = \frac{2N}{2\pi} = \frac{N}{\pi}$

Assuming that the signal is sampled at Nyquist rate, $f_s = \frac{2N}{\pi}$

$$\therefore \Delta \geq \frac{17N}{2N/\pi} \Rightarrow \Delta \geq 8.5\pi$$

(b) Prefiltered signal-to-quantization noise ratio:

$$\text{Maximum} (SNR)_Q = \frac{\text{Signal Power}}{\text{Quant Noise Power}} = \frac{\text{Signal Power}}{\Delta^2/3} = \frac{\frac{3^2}{2} + \frac{4^2}{2} + \frac{4^2}{2}}{(8.5\pi)^2/3}$$

$$= 0.0862$$